

NICOLAS SERRANO

Full Stack Software Engineer

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SUMMARY

Senior Software Engineer with 7+ years of experience building systems including infrastructure and systems at the Ethereum Foundation (PSE). Led development of the KZG Ceremony frontend, p0tion, MACI, with expertise in Typescript, Solidity, React, protocol architecture, cryptography, and production infrastructure.

SKILLS

Full-Stack: Typescript, React, Node (NestJS), Go (Gin), Rust (Axum - WASM), Rest API, Tailwind, Python, PostgreSQL, PostGIS, MongoDB, MySQL, Redis, WebAuthn passkeys, Progressive Web Apps (PWA).

Contracts: Solidity, Ethers, Viem, Hardhat, Foundry, IPFS, GraphQL, go-ethereum, ERC-20 / 721 / 4337.

Ethereum: L1/L2s, EVM, RLP encoding, MPTs, home validator (Linux, Ethrex + Lighthouse from source)

DevOps: AWS suite (S3, EC2, Lambda, Amplify, etc), Docker, GitHub Actions (CI/CD), TheGraph, Firebase

Cryptography: ZK (SNARKs), HE/FHE, MPC, zkWormholes, Post-Quantum Cryptography, Circom, Noir.

Agentic Engineering: Claude Code, Codex, hooks, spec-kit, test-driven planning, Ralph loops, Oh My Claude.

EXPERIENCE

Software & ZK Engineer | Ethereum Foundation — Privacy & Scaling Explorations (PSE)

July 2022 – July 2026 (4 years) | Remote

Trusted Setups Team | Staff Software Engineer

- Led frontend development for the **KZG Ceremony** (ceremony.ethereum.org), critical component to enable Danksharding (EIP-4844), **achieving 141,416 contributors** globally over 208 days. The **largest multi-party computation of its kind** at time of publication. Stack used: **Typescript, React, Ethers**, KZG library wrapper in **Rust** for efficient client-side secret generation, **IPFS** and **PWA** for censorship resistance, **ENS domains**. Deployed on **AWS** (Amplify and Cloud Monitoring) and **IPFS Pinata**.
- Co-built and maintained the **production p0tion platform**: the most widely used zkSNARK trusted setup framework in Ethereum. Owned development and maintenance for 2 years (backend, frontend and CLI). The platform **hosted 20+ finalized ceremonies** and **16,000 users** (ceremony.pse.dev). Stack used: **Typescript**, CommanderJS, **Ethers**, **SnarkJS**, **Firebase**, Google Cloud functions, **AWS S3** storage.
- Engineering led** on [brebaje](#), a p0tion-rewrite for multiple ecosystems (Ethereum, Cardano, etc). Advised Wonderland on [cabure](#), a p0tion-equivalent tool. Stack: **Typescript, React, NestJS, SQLite**, Speckit.

Private Transfers Team | Software Engineer and Researcher

- Led system design, metrics collection, frontend development, and evaluation review for a comprehensive private transfer protocols report on Ethereum: **55 protocols evaluated across architectures** (UTXOs, Stealth Addresses, Mixers, Private L2s, FHE, TEEs, etc.) and **35 properties** (including CROPS, compliance, etc). Metrics with **TheGraph** or **RPC nodes**, frontend with **React** and **Tailwind**, evaluations with **Claude** citations and human final deep review: <https://private-transfers.pse.dev/>
- Designed and implemented a zkWormholes system (**Private Proof of Burn**) with plausible deniability and client-side proving as invariants. Explored efficient client-side proving approaches by optimizing RLP encodings and sha256 in circuits. Used the Superchain **ERC20** framework developed for the OP ecosystem (**Solidity, Foundry, Typescript, React, Vite, Wagmi, Circom, Circomkit, SnarkJS**).
- Developed 2 post quantum private transfers systems. One based on **FHE + ORAM** (confidential amounts) and another **UTXO-based shielded pool**. Stack used: **Typescript, NextJS, React, Solidity, OpenZeppelin** and **Hardhat**. AI tools: Ralph loops, Claude Code agents with custom skills and hooks.

Private Voting Team | Senior Software Engineer

- Core contributor to [MACI](#) (**+596 stars** private voting protocol proposed by Vitalik. Used by Gitcoin, Aragon & Devcon). Primary focus on the coordinator: a **TypeScript NestJS backend service** orchestrating voting and tallying processes (versioned API + Swagger docs). Used **ZeroDev** for gasless txs (ERC-4337). Contributed to **TypeScript** SDK, **Solidity** contracts (**Hardhat**), and **Circom** ZK circuits. Addressed vulnerabilities found in **HashCloak** audit.
- Led integration of MACI private voting plugin for **Aragon OSx: Solidity/Foundry** plugin with unit tests, **Typescript NextJS** frontend using Aragon templates + MACI SDK, and integration documentation. Onboarded and **mentored a team member** through the MACI codebase and the Aragon OSx code.
- Coordinated the [Gitcoin GG24 Privacy Round](#) in collaboration with the external Privote team. Supported development (MACI SDK) and reviewed applications. **Round raised 35.41 WETH** in the voting contract.

Staff Smart Contract Engineer | CEDIA — National Research & Academic Consortium of Ecuador September 2021 – August 2022 (1 year) | Cuenca, Ecuador

- Built [e-voting system](#) used in **15 university elections (+150k voters)**. Owned on-chain component: private, decentralized & auditable. Stack: **Go/Gin, go-ethereum, Solidity, IPFS, PWA, Docker**.
- Provisioned **authenticated RPC** infra for Ethereum, Avalanche and Polygon on Linux VMs with Nginx.

Forward Deployed Engineer | Innwork S.A.S.

August 2020 – Present (on the side) | Ecuador

- Lead **team of 3-5** (developers, product managers and designers) to deliver high-quality solutions for clients pain points. Talked and negotiated with customers. Acted as project manager in multiple projects. Owned **15+ websites & e-commerce development**. Stack: Wordpress, PHP, MySQL, Stripe, Paypal.

Tech Lead & Co-Founder | Innomaps — Geoanalytics Startup

December 2017 – August 2022 (4 years 8 months) | Ecuador

- Co-founded [geoanalytics startup](#) recognized as Top 35 of 500+ projects by the [Hult Prize](#); grew to **2-5 team members, 5+ active corporate clients, 50+ monthly users, \$40,000 in revenue**. Stack: **Express, PostgreSQL, PWA, Python ML pipeline, Google ML, Google Maps API** integration.

EDUCATION

Information Technology Engineering — Cum Laude | Yachay Tech University

2015 – 2021 | Urcuquí, Ecuador

Thesis: [Data Marketplaces, Blockchain & HE](#). **Paper:** [A P2P Private Data Marketplace](#) (13 citations).

Certifications: Zero Knowledge Proofs MOOC (Stanford & UC Berkeley - 2023), Blockchain: Disruptive Technology (MIT - 2022), Startup Accelerator Program (Hult Prize Foundation - 2019)

PUBLICATIONS & RESEARCH

- [Auditor viewing keys in privacy systems](#). Personal blog, July 2026
- [Privacy in Lending Protocols](#). Personal blog, July 2026
- [State of Private Voting 2026](#). PSE Blog, November 2025
- [Learnings from the KZG Ceremony](#). PSE Blog, 2023
- [Blockchain: Sustainable Development or Energy Consumption?](#) Bionatura Journal, 2018. 5 citations.

COMMUNITY & TALKS

- [IC3 Blockchain Summer Camp](#) — 6-year consecutive participant and EF speaker alongside Arbitrum, Chainlink, JP Morgan. Given talks in Cornell University, Cornell Tech and Princeton University. [IC3 Hackathon 2022 first place winner team](#)
- Speaker at: [Encryption Day \(ARG, 25\)](#), [ProgCrypto \(IST, 23\)](#), [Aleph ZK Week \(ARG 24\)](#).
- [ETH Ecuador](#) — active member (15 people) team organizing courses, technical projects, and events.